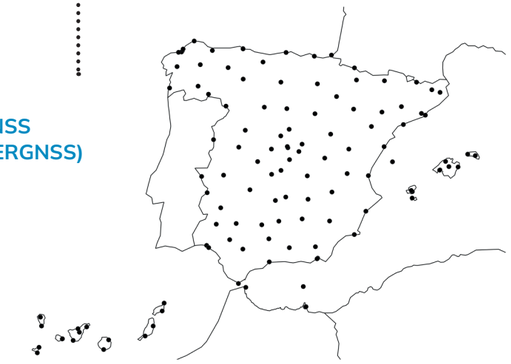
Permanent GNSS Stations

1 IGN Permanent GNSS Station Network (ERGNSS)

Since 1999 the National Geographic Institute has been establishing, expanding and computing a network of permanent GNSS stations (ERGNSS) distributed across the whole country.

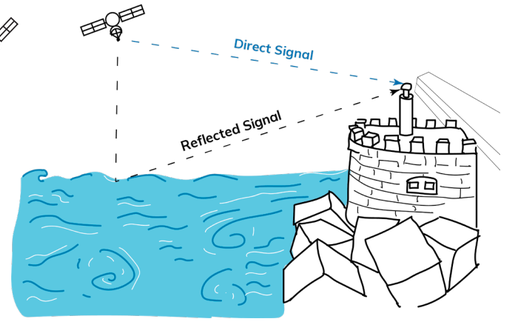




2 GNSS and tide gauges

A GNSS station is of great value for studying sea level: first, by computing the coordinates of the site, it can detect possible movements affecting the tide gauge and the port, which could otherwise degrade the sea-level record.

Secondly, using the GNSS reflectometry technique, sea level and its variation over time can be measured directly from satellite signals that are reflected off the water and recorded by the GNSS antenna.



Unofficial English translation prepared for the EAASI Partners Summit 2026. Original Spanish brochure © O. A. Centro Nacional de Información Geográfica (IGN-CNIG).

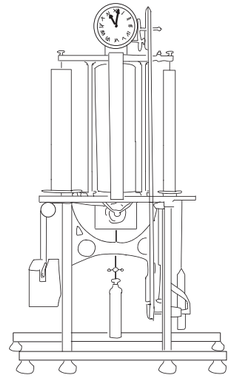
Fundamental STATION OF THE Mediterranean

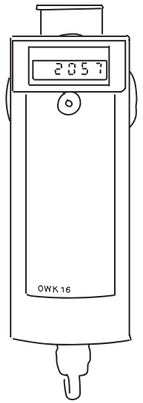
1 1874 - 1925

When the tide-staff observations ended in 1874, Spain's first tide gauge was put into operation in the port's Public Works building: a float gauge designed by Mr Adie, which recorded on paper the movement of a float on the water inside a well connected to the sea.

In 1914 it was moved to the Harbour Pilots' Pavilion, where it kept recording until 1924.

In 1925 construction began on the Fundamental Station of the Mediterranean, now known as Alicante 1 (ALAC1). It was designed as part of a network of tide gauges along the country's entire coastline.





2 Today

Over almost a hundred years of data recording at this tide gauge station, many different instruments have been used: from the oldest mechanical paper recorders, such as the Thomson-type tide gauge, to punched-tape recorders with coded output. Today only digital sensors operate here (angular encoders and radar sensors), sending their measurements in real time to the headquarters of the National Geographic Institute.

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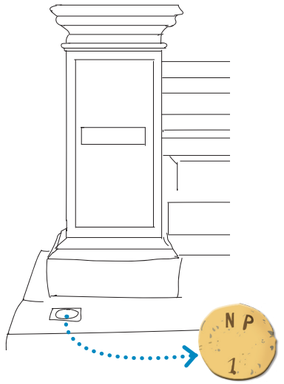
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MINISTERIO DE TRANSPORTES Y MOVILIDAD SOSTENIBLE
INSTITUTO GEOGRÁFICO NACIONAL

Tide Gauge AND GEODETIC STATION OF Alicante



National HEIGHT Reference



1 Tide staff



Producing the National Topographic Map required a levelling network covering the whole country and an origin for heights.

Between 1870 and 1874, daily sea-level readings were taken on a tide staff located in the port of Alicante.

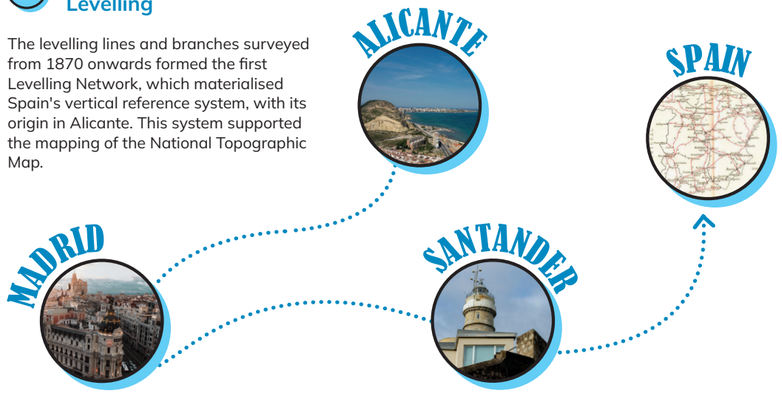
2 National Height Reference

Averaging the observations gave the Mean Sea Level, 0.430 m above mark Z, on which the zero of the staff rested.

Mark NP1, a bronze disc set in the first step of the main entrance to Alicante City Hall, was adopted as the National Height Reference. Precise levelling between marks Z and NP1 established the height of NP1 at 3.4095 m above Mean Sea Level in Alicante (NMMA) in 1874.

3 Precise Levelling

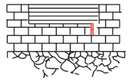
The levelling lines and branches surveyed from 1870 onwards formed the first Levelling Network, which materialised Spain's vertical reference system, with its origin in Alicante. This system supported the mapping of the National Topographic Map.



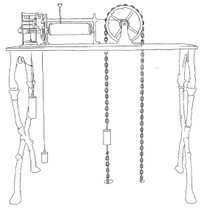
MEASURING Devices

1 Tide staff

Metal staff graduated in centimetres, read directly by eye.



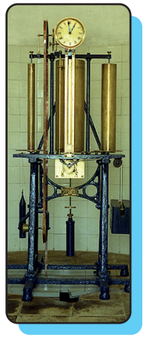
2 Float Gauge



Mechanical recorder that captures the movement of the sea through a float placed in a well connected to the open sea.

The float hangs from a chain or tape that runs over a wheel driving a pen. The pen draws on a sheet of paper wrapped around a cylinder, which in turn is rotated by a clockwork mechanism.

Over time, pendulum clockwork and continuous paper rolls were introduced, allowing records of up to a month. These innovations brought greater precision and much more practical handling.



In the 1960s automatic recorders came into use. They captured the movement of the sea in a similar way to the earlier ones, but stored the data on punched paper tape with binary coding.



3 Angular Encoder



Angular encoders still capture the movement of the sea through a float in the water.

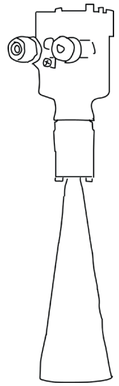
The wheel turned by the float via a tape converts changes in level into rotary movements, which are translated directly into a digital signal by a contactless magnetic sensor.

The system includes a memory where the data are stored. Originally, data were downloaded directly by telephone. Today the system downloads automatically, so the data reach the IGN in real time.

4 Radar Sensor

Radar sensors work without any contact with the water.

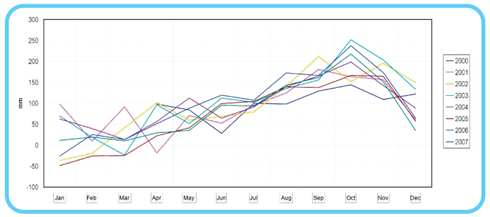
The antenna emits short radar pulses that reach the water, are reflected by it and are picked up by the antenna as an echo. The elapsed time is proportional to the distance, which gives the water level, stored in digital form.



TIDES IN Alicante

1 Tide in Alicante

The tide is the periodic rise and fall of sea level. It is mainly caused by gravitational forces and by the distances, declinations and relative movements of the Earth, the Moon and the Sun. The effects of these bodies can be represented as waves, whose combination gives the theoretical tide.



Seasonal variability



But meteorological factors (atmospheric pressure, wind, water inflow...) and physical factors (water depth, seabed relief, salinity...) also affect the tide.

A change of 1 mbar in pressure shifts sea level by about 1 cm in the opposite direction.

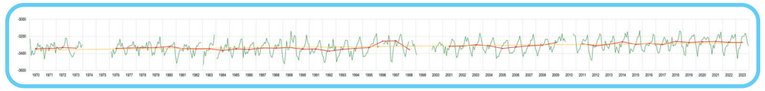
The tide shows seasonal variability due to atmospheric pressure, wind and water temperature, which is warmer in summer and autumn.

2 Observed Tide

Today, digital readings are taken every minute.

Hourly data are used to calculate daily, monthly and annual means.

A long series of annual means makes it possible to derive a sea-level trend.



Annual means